Route Resilience System

Machine Learning Project Report

Urban Road Network Resilience Analysis using
Deep Learning, Graph Theory & Geospatial Intelligence

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Key Technologies

Python · PyTorch · U-Net · NetworkX · FastAPI · Next.js · OSMnx

A Machine Learning approach to making cities safer and more resilient

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1. Introduction — What Is This Project?

The **Route Resilience System** is a **Machine Learning project** that analyses urban road networks to find out **how strong or weak** they are against disasters like earthquakes, floods, and landslides.

Think of a city's road network like a spider web. If you cut one strand in the middle, the whole web becomes weak. Similarly, if a critical road junction in a city gets damaged, thousands of people could get stuck — ambulances can't reach hospitals, fire trucks can't reach fires, and daily commuters face hours of delay.

This project uses **Deep Learning (U-Net)** to extract road networks from satellite images, converts them into mathematical graphs, and then uses **graph theory algorithms** to identify the most critical junctions. It then simulates disasters to show what would happen if those junctions fail — and suggests fixes.

In Simple Terms: This ML system looks at satellite images of a city, understands the road network, finds the weak points, simulates what happens when disasters strike, and tells the government exactly where to invest to make the city safer.

Key Objectives

- **Extract** road networks from satellite imagery using deep learning (U-Net).
- **Build** a mathematical graph of the city's roads (nodes = intersections, edges = roads).
- **Identify** critical junctions whose failure would cause maximum disruption.
- **Simulate** natural disasters and measure their impact on the road network.
- **Recommend** where to build new roads or repair existing ones for maximum resilience.
- **Visualise** everything on an interactive web dashboard for easy decision-making.

2. Problem Statement — Why Do We Need This?

India's cities are growing rapidly. Roads that were built decades ago now carry 10× more traffic. When a natural disaster strikes — an earthquake in Gujarat, flooding in Chennai, or landslides in Uttarakhand — the road network is often the first thing to fail.

> *"In the 2015 Chennai floods, over 500 roads were submerged and hundreds of thousands of people were stranded for days. Emergency vehicles couldn't reach critical areas because nobody knew which alternative routes were still functional."*

The core problem:

- Cities don't know which road junctions are **most critical** to the entire network.
- There is no way to **simulate disasters in advance** and predict road failures.
- Government decisions on road repair and new construction are often based on **political pressure** rather than **data-driven analysis**.
- During disasters, there is no tool to **quickly identify alternative routes** and plan evacuations.

This project solves all of these problems using **Machine Learning and Graph Theory**.

Problem Statement Context

This project addresses **ISRO NNRMS Hackathon 2026 — Problem Statement PS4**: *"Urban Road Network Resilience Analysis using Satellite Imagery and Graph Theory."* The challenge requires extracting road networks from satellite data, constructing weighted graphs, computing centrality metrics, and simulating disruptions to assess network resilience.

3. System Architecture

The diagram below shows the complete system architecture — from raw data input to the final dashboard used by citizens and government officials.

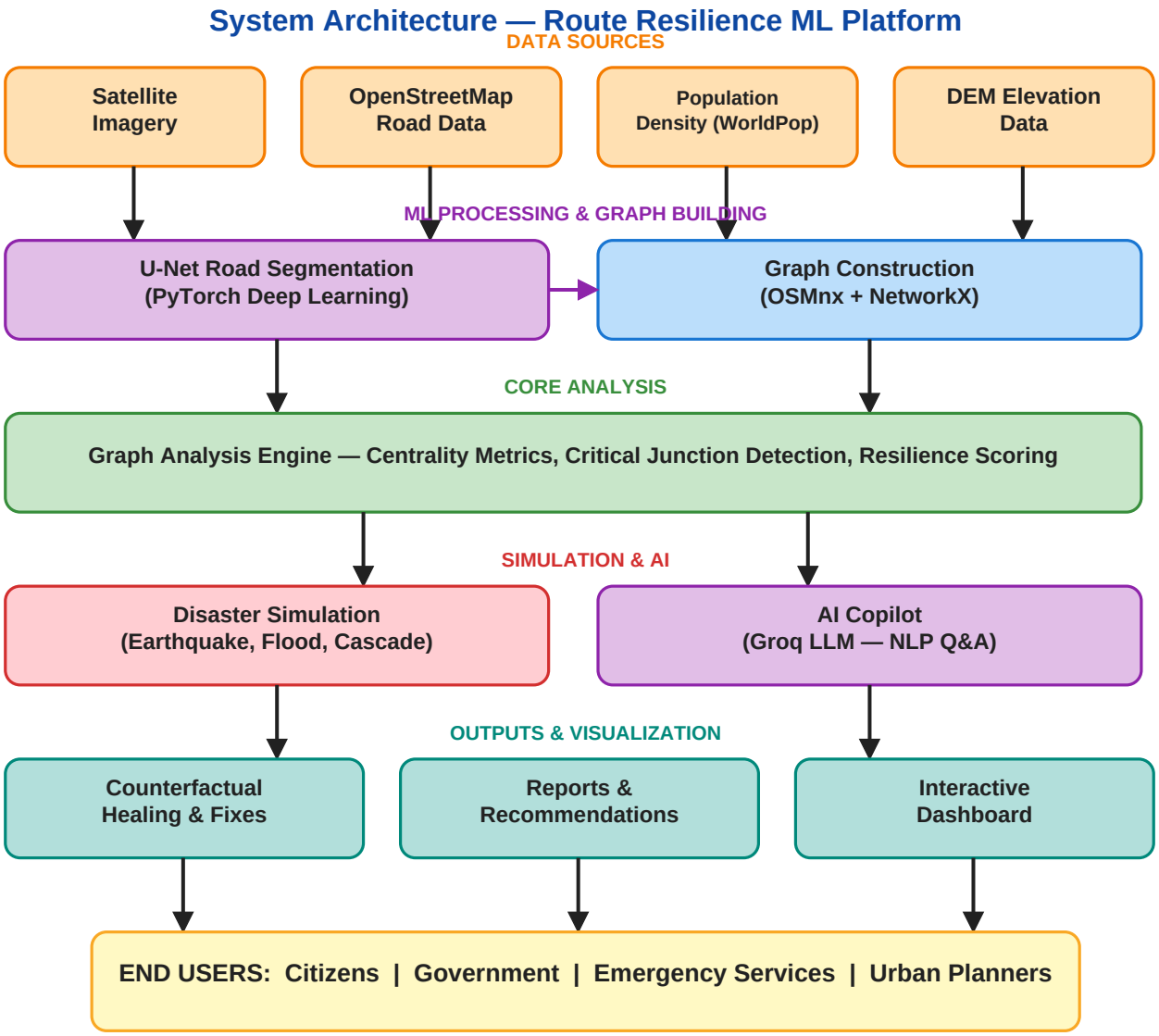

Figure 1: Complete System Architecture of the Route Resilience ML Platform

The system is divided into **5 layers**:

- **Data Sources Layer:** Satellite images, OSM road data, population density, elevation models.
- **ML Processing Layer:** U-Net deep learning model extracts roads from images; OSMnx builds graphs.
- **Analysis Layer:** Graph algorithms compute centrality, detect critical junctions, score resilience.
- **Simulation & AI Layer:** Disaster simulations + AI Copilot for natural-language insights.

• **Output Layer:** Interactive dashboard, auto-generated reports, and healing recommendations.

4. Machine Learning Pipeline

The ML pipeline is the core intelligence of the system. Here is the step-by-step flowchart showing how a satellite image gets transformed into actionable road network insights.

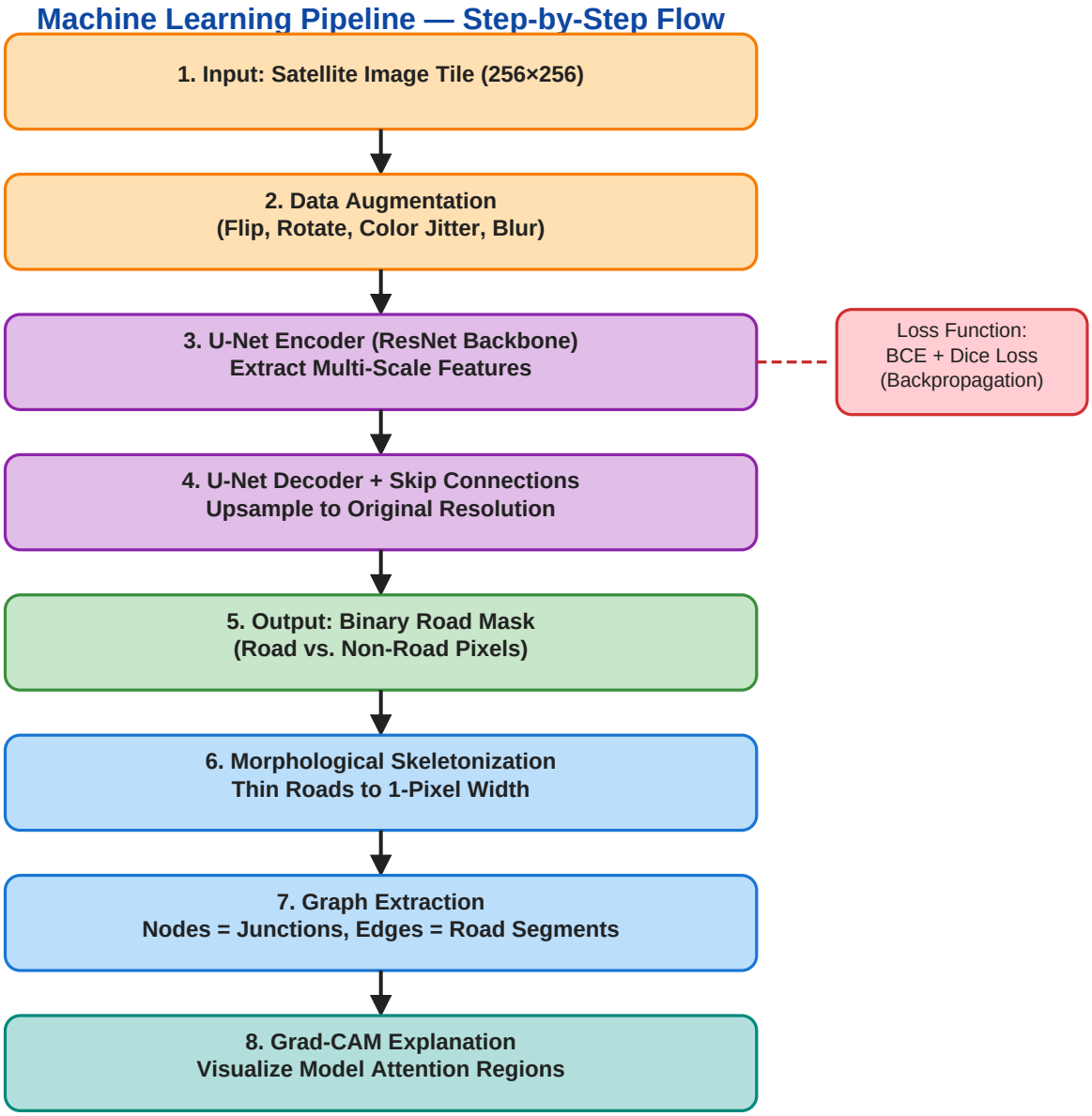

Figure 2: Machine Learning Pipeline — From Satellite Image to Road Graph

How it works in simple words: A satellite photo of the city goes in → the ML model (U-Net) identifies which pixels are roads → those road pixels are thinned to single lines → the lines are converted to a mathematical graph → the graph is analysed for weak points.

5. U-Net Architecture

The U-Net is a **convolutional neural network** specifically designed for image segmentation. It has an **encoder** (downsampling path) that captures context and a **decoder** (upsampling path) that enables precise localisation. **Skip connections** bridge the two paths to preserve fine details.

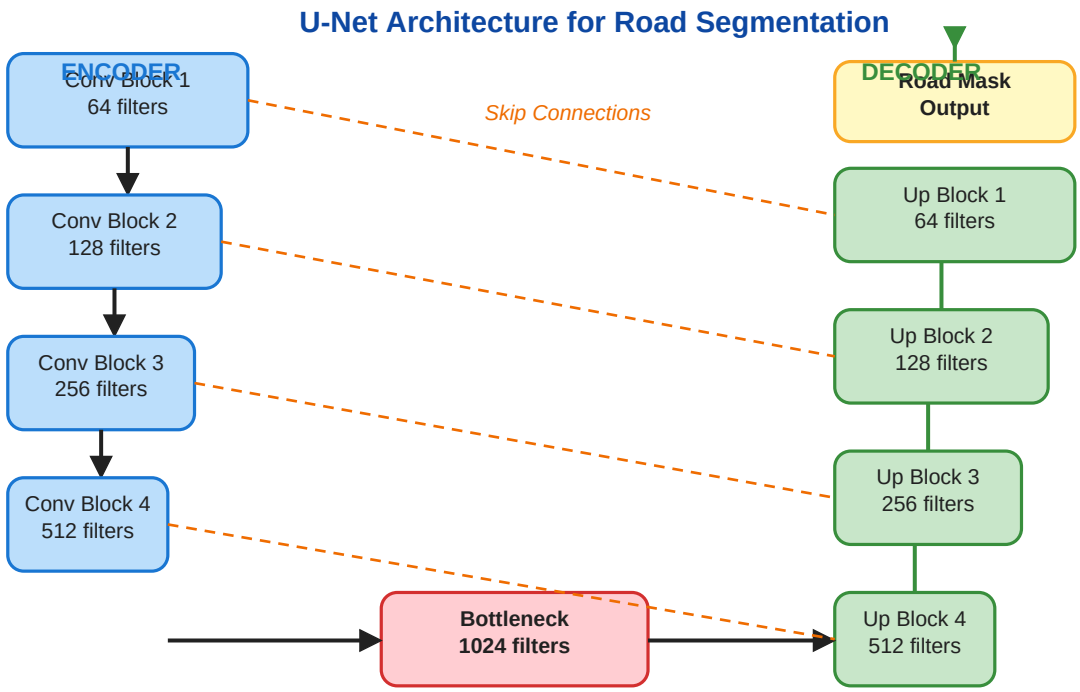

Figure 3: Simplified U-Net Architecture Used for Road Segmentation

Why U-Net for Road Segmentation?

- **Pixel-level accuracy:** U-Net classifies every pixel as road or non-road.
- **Skip connections:** Preserve fine details like narrow roads and sharp turns.
- **Works with limited data:** Originally designed for biomedical images with few samples.
- **Fast inference:** Processes a 256×256 tile in milliseconds.

Training Details

Parameter	Value
Input Size	256 × 256 × 3 (RGB satellite tile)
Output	256 × 256 × 1 (binary mask: road / non-road)
Encoder Backbone	ResNet (pretrained on ImageNet)
Loss Function	Binary Cross-Entropy + Dice Loss

Parameter	Value
Optimizer	Adam (lr = 1e-4)
Augmentations	Random flip, rotation, colour jitter, Gaussian blur
Framework	PyTorch + segmentation-models-pytorch

6. How It Works — Step by Step

Here is a simple, easy-to-follow explanation of how the system works:

Step	What Happens	Technology Used
1	Satellite images of the target city are obtained (or road data is fetched from OpenStreetMap).	OSMnx, OpenStreetMap
2	The U-Net ML model scans the satellite images and identifies road pixels — like highlighting roads with a marker.	PyTorch, U-Net CNN
3	Road pixels are 'skeletonized' — thinned to single-pixel-wide lines so the computer can trace each road.	Morphological Thinning
4	The road skeleton is converted into a mathematical graph: every intersection becomes a 'node', every road between intersect	NetworkX
5	The system calculates how important each junction is using centrality measures (betweenness, closeness, degree, eigenvec	NetworkX, SciPy
6	Critical junctions are identified — these are the junctions whose failure would cause the most damage to the overall network.	Graph Theory
7	The user selects a disaster scenario (earthquake, flood, etc.) and the system removes affected junctions/roads from the grap	Simulation Engine
8	The system measures how badly the network is damaged: How many islands did it break into? How much longer are routes n	Resilience Metrics
9	The system suggests fixes: 'Build a bypass road here' or 'Repair this bridge first' — each with a cost-benefit score.	MST Healing Algorithm
10	Everything is displayed on a beautiful interactive dashboard with maps, charts, and downloadable reports.	Next.js, Leaflet, Recharts

7. Key ML Techniques Used

Technique	Category	How It's Used
U-Net (CNN)	Semantic Segmentation	Classifies every pixel in a satellite image as 'road' or 'not road'. The encoder-decoder arc
Transfer Learning	Pre-trained Backbone	The encoder uses a ResNet backbone pre-trained on ImageNet, which provides robust f
Grad-CAM	Model Explainability	Gradient-weighted Class Activation Mapping generates heatmaps showing which image
Change Detection	Temporal Analysis	Compares road segmentation masks from two time periods to detect new roads or dama
Graph Neural Concepts	Centrality Analysis	While not a GNN, the project applies graph-theoretic metrics (betweenness, eigenvector
NLP (LLM)	AI Copilot	Integrates Groq's LLM (Llama/Mixtral) for natural-language Q&A about the road network.
Data Augmentation	Training Robustness	Random flips, rotations, colour jitter, and Gaussian blur prevent overfitting and make the

8. Benefits to Citizens & Society

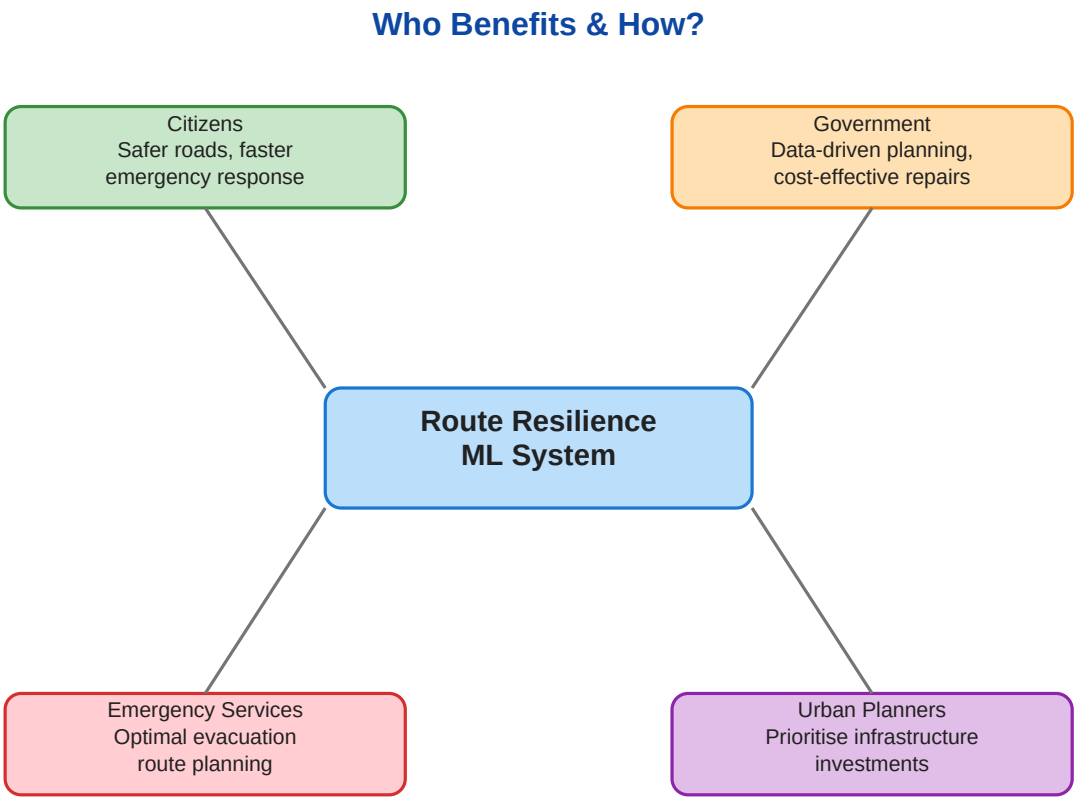

Figure 4: Stakeholders Who Benefit from the Route Resilience System

How Citizens Benefit

• **Safer commutes:** The system identifies dangerous junctions that could fail. Once fixed, daily commuters face fewer risks.

• **Faster emergency response:** During disasters, evacuation routes are pre-computed. Ambulances, fire trucks, and rescue teams know exactly which roads are still functional.

• **Less disruption:** By predicting which roads will fail during floods or earthquakes, the government can proactively strengthen them — meaning fewer road closures and detours.

• **Equity in infrastructure:** The equity analysis module ensures that poorer neighbourhoods (which are often ignored) get fair attention in road resilience planning.

• **Real-time information:** The interactive dashboard can show citizens which roads are safe and which are damaged, helping them plan their routes during emergencies.

Social Impact

In India, road accidents and disaster-related road failures cause thousands of deaths annually. A system that can predict weak points and suggest fixes **before** a disaster strikes can save lives. The population impact analysis also ensures that areas with the highest population density get priority attention.

9. Benefits to Government & Administration

9.1 Data-Driven Decision Making

Currently, most road repair and construction decisions are made based on complaints, political pressure, or visual inspection. This system provides **quantitative, data-driven evidence** for every decision:

• "Junction X has a betweenness centrality of 0.47 — if it fails, 23% of all shortest paths in the city are disrupted."
• "Building a bypass road between Node A and Node B would improve network resilience by 18% at a cost of only 2.3 km of new road."

9.2 Cost Optimization

Government budgets for road infrastructure are limited. This system helps allocate funds where they matter most:

• **Prioritised repair lists:** Instead of fixing roads randomly, the system ranks repairs by impact on overall network resilience.
• **Cost-benefit analysis:** Every proposed intervention comes with a resilience gain score vs. construction cost.
• **Avoid over-investment:** Some roads look important but aren't critical to the network. The system identifies them, saving unnecessary spending.

9.3 Disaster Preparedness

• **Pre-disaster planning:** Simulate earthquake/flood scenarios and prepare response plans before the disaster actually hits.
• **Evacuation route pre-computation:** Know in advance which routes to use for evacuation under different disaster scenarios.
• **NDMA integration potential:** Can feed directly into the National Disaster Management Authority's planning systems.

9.4 Urban Planning

• Identify areas where the road network is **too sparse** and needs new connections.
• Plan **smart city** road networks with built-in resilience.
• Evaluate the impact of proposed new roads/flyovers before construction begins.

10. Benefits to Emergency Services & Disaster Management

During a Disaster

• **Real-time network status:** Instantly see which roads are damaged and which are still functional.

• **Optimal evacuation routes:** The evacuation module computes the fastest routes from danger zones to safety, considering road capacity and damage.

• **Resource allocation:** Know which areas are completely cut off and need helicopter/boat rescue vs. those reachable by road.

Before a Disaster

• **Vulnerability mapping:** Identify which neighbourhoods would be isolated if specific roads fail.

• **Drill planning:** Use simulation results to design realistic disaster response drills.

• **Strategic positioning:** Position fire stations, ambulance depots, and relief centres at locations that remain accessible even when critical junctions fail.

After a Disaster

• **Prioritised recovery:** Which roads should be cleared first to restore maximum connectivity?

• **Damage assessment:** Change detection (ML) compares pre-disaster and post-disaster satellite images to automatically map road damage.

• **Cascade prevention:** The cascade failure model shows which damaged roads could trigger further failures — helping teams prevent secondary disasters.

11. Economic & Long-Term Benefits

Benefit Area	Description
Reduced Disaster Losses	By strengthening critical junctions before disasters, the economic losses from road network failures ca
Lower Repair Costs	Proactive maintenance (fixing weak points early) is 3-5× cheaper than emergency repairs after a disa
Fuel & Time Savings	When critical junctions fail, commuters take long detours. The system's recommendations reduce ave
Insurance Benefits	Cities with quantified resilience scores may qualify for lower disaster insurance premiums.
Tourism & Business	A resilient road network improves a city's reputation, attracting business investment and tourism.
Job Creation	Building recommended bypass roads and bridges creates construction jobs while improving the city's
Healthcare Access	Ensuring hospitals remain reachable during disasters reduces mortality and healthcare costs.

12. Real-World Use Cases

Use Case	How the System Helps
Chennai Flood Preparedness	Simulate monsoon flooding by disabling low-elevation roads. Identify which hospitals, schools, and resident
Earthquake Response (Gujarat)	Simulate a magnitude 7.0 earthquake with a configurable epicentre. The cascade model shows how infrastr
Smart City Planning	Before building a new smart city, simulate the proposed road network. Test its resilience by removing junctio
Highway Authority (NHAI)	NHAI can use the system to identify critical segments of national highways. Prioritise maintenance budgets
Military | Defence	Identify strategically critical road junctions for national security. Simulate enemy attacks on key infrastructur
Post-Disaster Damage Survey	After a landslide, upload post-disaster satellite images. The change detection ML model automatically ident

13. Technology Stack

Category	Technology
Machine Learning	PyTorch, torchvision, segmentation-models-pytorch
Deep Learning Model	U-Net with ResNet encoder backbone
Explainability	Grad-CAM (Gradient-weighted Class Activation Mapping)
Graph Analysis	NetworkX (centrality, MST, shortest paths, components)
Geospatial	OSMnx, GeoPandas, Shapely, pyproj, rasterio
Backend API	FastAPI (Python) + Uvicorn ASGI server
Frontend	Next.js 14, React 18, TypeScript
UI Components	Tailwind CSS, Radix UI, shadcn/ui
Maps	Leaflet + react-leaflet (interactive map visualization)
Charts	Recharts (data visualization library)
LLM / NLP	Groq API (Llama / Mixtral for AI Copilot)
Data Sources	OpenStreetMap, ISRO Bhuvan, WorldPop, SRTM DEM

14. Data Flow

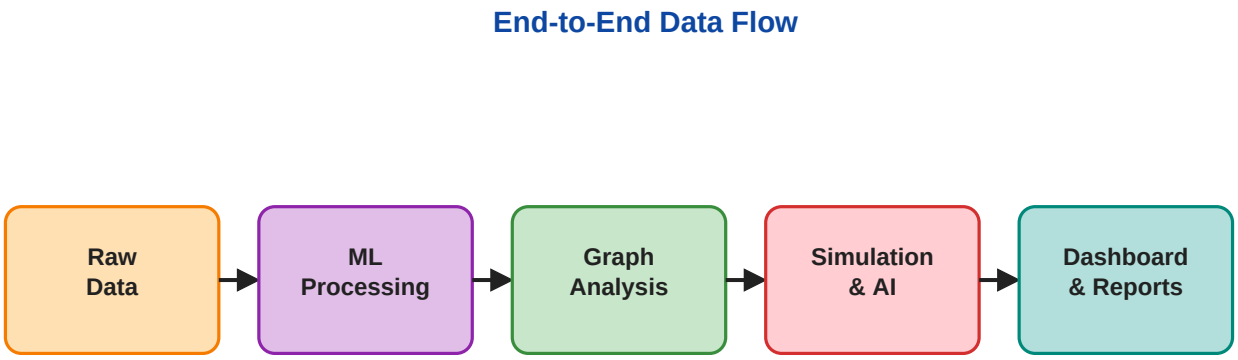

Figure 5: End-to-End Data Flow Through the System

Detailed data flow:

- **Raw Data** → Satellite images (GeoTIFF) + OSM road data (XML/JSON) + Population rasters + DEM tiles
- **ML Processing** → U-Net segments road pixels → Skeletonize → Extract graph topology
- **Graph Analysis** → Compute centrality metrics → Rank critical junctions → Calculate resilience score

- **Simulation & AI** → Run disaster scenarios → Measure fragmentation → AI Copilot answers questions
- **Dashboard & Reports** → Interactive maps + Charts + PDF reports + Healing recommendations

15. Results & Outputs

15.1 Graph Statistics (Bengaluru AOI)

Metric	Value
Total Nodes (Intersections)	13,486
Total Edges (Road Segments)	19,117
Average Node Degree	~2.84
Graph Type	MultiDiGraph (directed, weighted)
Coordinate System	WGS 84 (EPSG:4326)

15.2 System Outputs

Output	Description
Criticality Heatmap	Interactive map showing colour-coded junctions (red = most critical, green = least critical). Helps visually
Resilience Score	A single number (0–1) representing overall network health. Drops after disasters, rises after interventions
Disaster Impact Report	Before/after comparison of all metrics: efficiency drop, new connected components, average detour incre
Healing Recommendations	Ranked list of proposed new road segments with cost (km) and resilience gain (Δ efficiency). Actionable f
Evacuation Routes	Computed safest paths from danger zones to shelters, considering road capacity and predicted damage.
AI Copilot Responses	Natural-language answers to questions about the network, powered by Groq LLM with injected graph cor
PDF/HTML Reports	Downloadable reports summarizing all findings — suitable for presentations to government officials.
Change Detection Maps	Before/after satellite image comparison showing where roads were damaged or newly built.

16. Future Scope

The project has significant potential for expansion:

Enhancement	Description
Real-Time Traffic Integration	Connect to Google Maps / HERE traffic APIs to add real-time traffic flow as edge weights. Makes
Multi-City Support	Currently focused on Bengaluru. Can be extended to any city worldwide with a city-selector UI an
Live Satellite Monitoring	Integrate with satellite data providers for near-real-time damage detection after disasters using th
Mobile App	Build a lightweight mobile app for field teams to report road damage and view evacuation routes c
GNN Integration	Replace classical centrality with Graph Neural Network-based importance scoring for more nuanc
NDMA / SDMA Integration	Direct API integration with National/State Disaster Management Authority systems for automated
Historical Replay	Load past disaster data and validate model predictions against actual damage — improving mode
Docker Deployment	Containerise the entire system for one-command deployment on cloud platforms (AWS, GCP, Az
Reinforcement Learning	Use RL to optimize intervention strategies: which combination of new roads gives the best resilien

17. Conclusion

The **Route Resilience System** demonstrates how **Machine Learning** can be applied to a real-world problem that affects millions of people. By combining **deep learning (U-Net)** for road extraction, **graph theory** for network analysis, and **simulation algorithms** for disaster modeling, the system provides a complete, end-to-end solution for urban road network resilience.

The system is not just a theoretical exercise — it produces **actionable insights** that governments can use to save lives and money. From identifying the single junction whose failure would cause the most damage, to recommending exactly where a new bypass road should be built, every output is designed to be **practical and decision-ready**.

With its **interactive dashboard**, **AI Copilot**, and **automated report generation**, the system makes complex ML and graph theory results accessible to non-technical users — city planners, disaster management officials, and even citizens.

Key Takeaway: This ML project proves that Deep Learning + Graph Theory can make cities safer, save government money, speed up emergency response, and ultimately save lives — all from satellite images and open-source road data.

— End of ML Project Report —